Mid-term Test

Summary

- **2 November, 15 :15**, 1h30 (+30 min special arrangements), assigned room and seat, CAMIPRO on the table.
- Paper, 12 pages; **one A4 sheet** (recto-verso, free format); no electronic device; black or dark blue pen; SCIPER on top.
- Single-choice + short answers : concise but full sentences, **1 point** ≈ **1-2 sentences**; French allowed.

Bias in Machine Learning

Definition

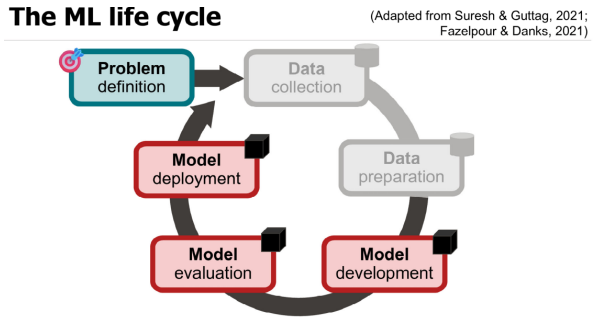
MACHINE LEARNING : techniques that **infer patterns from data** that **generalize to unseen data**. Algorithm = process (finds the parameters); model = result (function + parameters). Training, then inference $y_u = f(x_u)$.

Key formulas

Supervised (labels y) : regression (y continuous), classification (y discrete).
Unsupervised : clustering, dimensionality reduction.

Key idea

Sensitive to **data bias** (pre-existing, representation, measurement); labels often come from **human annotation**.



Key formulas

Life cycle biases

- **Optimization choices** : accuracy only ⇒ unfairness.
- **Aggregation bias** : different groups merged (Simpson's paradox).
- **Evaluation bias** : benchmark ≠ served population (Gender Shades, intersectional).
- **Deployment bias** : used differently than intended.

Example

Shoplifting detector tested on actors simulating thefts → **evaluation bias**.

Explained to a goat

The machine learns a rule from examples. Bad examples, a rule mixing different goats, a test on the wrong goats, or a use it was not made for : all give unfair answers.

Group Fairness Criteria

Definition

FAIRNESS METRICS : mathematical measures of *certain* notions of fairness. Compare the confusion matrices of **two groups**. (Criticisms : only equality, oversimplified, blind optimization.)

Key formulas

Actual values (ProPublica, **defendant**) :

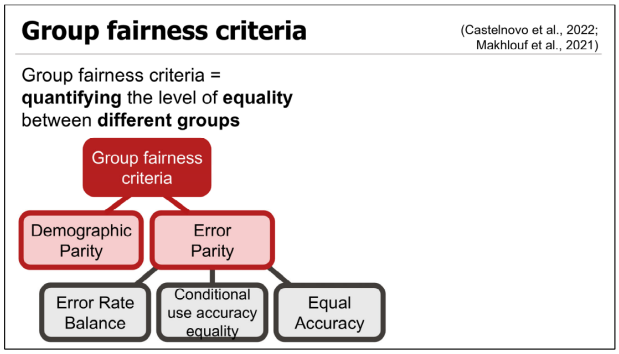
$$\text{FNR} = \frac{FN}{TP + FN} \qquad \text{FPR} = \frac{FP}{TN + FP}$$

Predicted values (NorthPointe, **decision-maker**) :

$$\text{FDR} = \frac{FP}{TP + FP} \qquad \text{FOR} = \frac{FN}{TN + FN}$$

Acceptance rate :

$$\frac{TP + FP}{TP + FP + FN + TN}$$



Definition

ERROR PARITY (outcome *quality*) : error rates independent of the group.

- **Error rate balance** : FPR, FNR (FNR only : equal opportunity).
- **Conditional use accuracy equality** : PPV/FDR, NPV/FOR (PPV only : predictive parity).
- **Equal accuracy**.

DEMOGRAPHIC PARITY (outcome *distribution*) : equal acceptance rates (statistical parity; disparate impact = ratio).

Warning

Impossibility : different base rates + imperfect classifier ⇒ the criteria cannot all hold. Choosing one is a **moral dilemma**.

Example

Fissures : plain $4/8$ vs. reinforced $3/9$ positive predictions → **demographic parity**, not satisfied. Every metric works for fairness if two groups are compared.

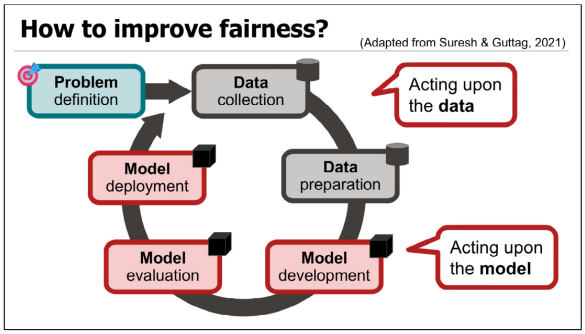
Explained to a goat

Build the robot's mistake table for each colour of goat and compare : among the guilty, among the accused, or how often each gets a "yes". You cannot equalise everything : choose.

Fairness Returns

Key formulas

Beyond groups : **intersectional** (combined groups; small samples), **individual** (similar people, similar decisions; similarity measure), **causal** (causal graphs; expert knowledge).
Improving : **pre-processing** (data : unawareness fails because of proxies, reweighting, resampling), **in-processing** (training objectives), **post-processing** (outputs; differential treatment).



Warning

Dilemmas : leveling down; fairness vs. accuracy; fairness vs. privacy (sensitive attributes needed).

Explained to a goat

Fix the data before, the learning during or the answers after; each fix costs something.

Strategy : Datasheets for Datasets

Key formulas

Datasheet (Gebru 2021) : motivation, composition, collection process, preprocessing, uses, distribution, maintenance. Reveals issues; engineers spot ethical issues **earlier and more often**.

Example

Face dataset. Safety : no consent (authors, website), re-identification, offensive content. Fairness : unclear population, no subgroup info, biased cropping. Model : unfit population, different error rates.
Red flags : "publicly available on the web", **no ethical review**.

Model cards : the same for models (intended use, factors, metrics, intersectional results, ethics).

Explained to a goat

A dataset needs an ID card; reading it shows the problems before your model learns them.

Strategy : People Behind the Data

Key formulas

IDEO AI Ethics Cards : **humanize** the data.

1. Find documentation, raw data.
2. Read the questionnaire, pick variables.
3. Pick participants, imagine their story.
4. Conclude on ethical impacts and risks.

Procedure and COMPAS example

COMPAS data from Propublica

Name	Age	Ethnicity	Risk of Violence	Risk of Recidivism	Two Years Recid
Taveunshae Livingston	20	African-American	10 – High	10 – High	No
Random high crime IQ criminal	32	Caucasian	Probably Low	Probably Low	Probably Yes

Example

COMPAS : a provocative young woman scores high risk but does not re-offend; a cunning criminal answers "safely" and scores low. The questionnaire captures an environment people did not choose (**pre-existing bias**) and oversimplifies lives (**measurement bias**).

Explained to a goat

Pick one row of the table and imagine the real goat behind it.

Weekly Summary

Summary

- ML flavours; aggregation, evaluation, deployment bias.
- FNR / FPR (actual, defendant) vs. FDR / FOR (predicted, decision-maker).
- Error parity (3 kinds) vs. demographic parity; impossibility.
- Intersectional, individual, causal fairness; pre-, in-, post-processing.
- Datasheets, model cards, people behind the data.